

From two columns to a testable chart choice

Same task. Actual pixels. One score that determines the ranking.

1 RAW OBSERVATIONS

A: benign B: malignant

0.2699	1.0950
0.1852	0.5435
0.2773	0.7456
0.1839	0.4956
0.1563	0.7572



2 RENDER AND RECOVER

21 settings • 7 eligible chart families

Pixels + calibrated axes → five differences

$$S = 100 / (1 + \text{normalized recovery loss})$$



3 SELECT

Heatmap

97.981 / 100

Best under this protocol

All 357 + 212 records analysed

Actual selected raster • reference resolution

A

B



-0.36 1.5 3.3 -0.36 1.5 3.3

Value • White → blue: bin probability 0 → 1

RANK / CHART AND SETTING

SCORE / 100

1	Heatmap 64 bins		97.981
2	Dot plot raw observations		96.956
3	Histogram 64 bins		96.686
4	Violin plot 32 grid points		92.868
5	Bar chart means		60.525
6	Interval plot five-number		52.309
7	Box plot five-number		52.136

Higher is better • Green: computational optimum

15-family catalogue screened; seven implemented for this task. Scores are computational, not human accuracy.

When changing the chart helps

Large numerical gains over mean bars

True difference Recovered from mean bars Recovered from selected chart

a Breast tumour / Fractal dimension (worst)

A: Benign (n=357)

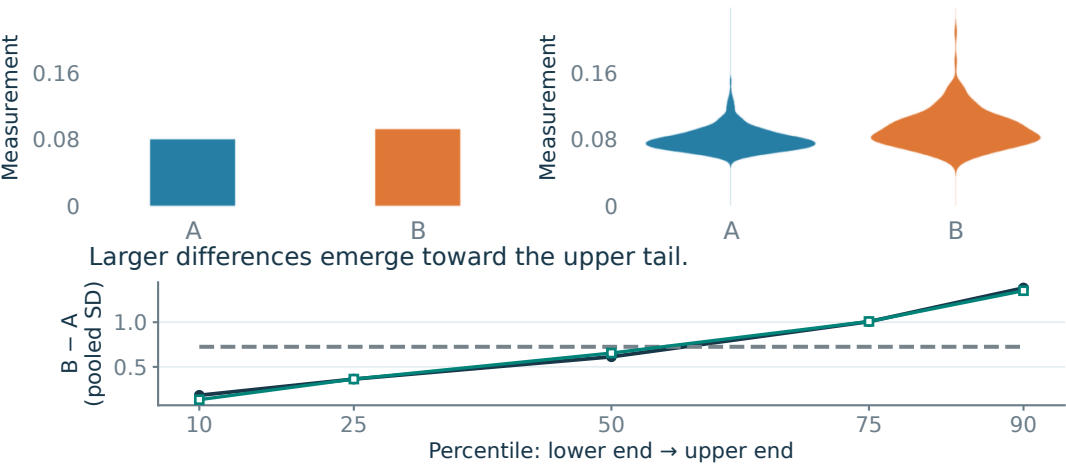
B: Malignant (n=212)

Mean bars 71.87

→ Violin plot 97.57

+25.70

score points



RANK / CHART		SCORE / 100	
1	Violin plot 256 grid points		97.566
2	Dot plot raw observations		96.724
3	Heatmap 32 bins		96.540
4	Histogram 32 bins		96.278
5	Bar chart means		71.865
6	Interval plot five-number		61.220
7	Box plot five-number		61.126

b Smartphone activity / Gyroscope-jerk variability

A: Sitting (n=30)

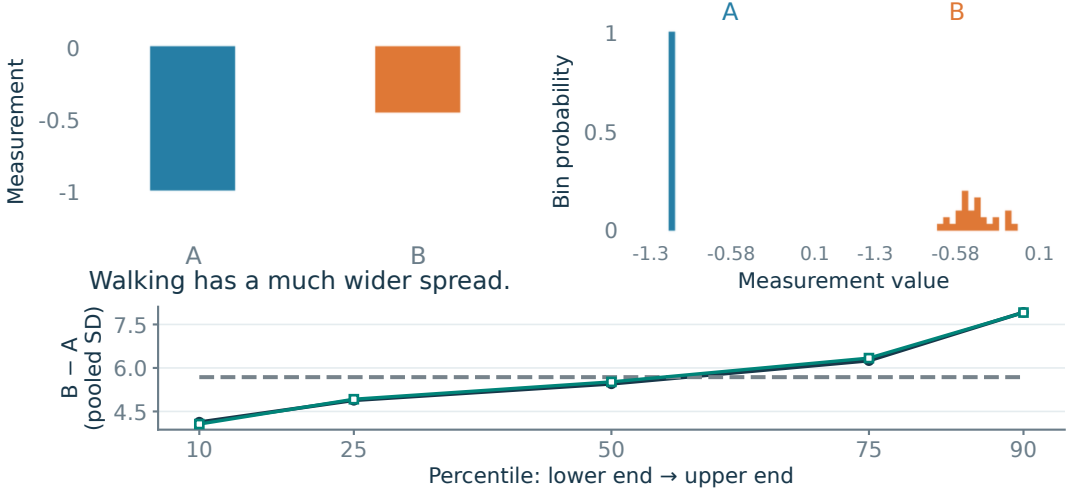
B: Walking (n=30)

Mean bars 48.15

→ Histogram 94.96

+46.81

score points



RANK / CHART		SCORE / 100	
1	Histogram 32 bins		94.960
2	Heatmap 32 bins		93.974
3	Box plot five-number		92.907
4	Dot plot raw observations		90.020
5	Interval plot five-number		89.943
6	Violin plot 32 grid points		87.802
7	Bar chart means		48.148

c Heart disease / ST depression (oldpeak)

A: No disease (n=164)

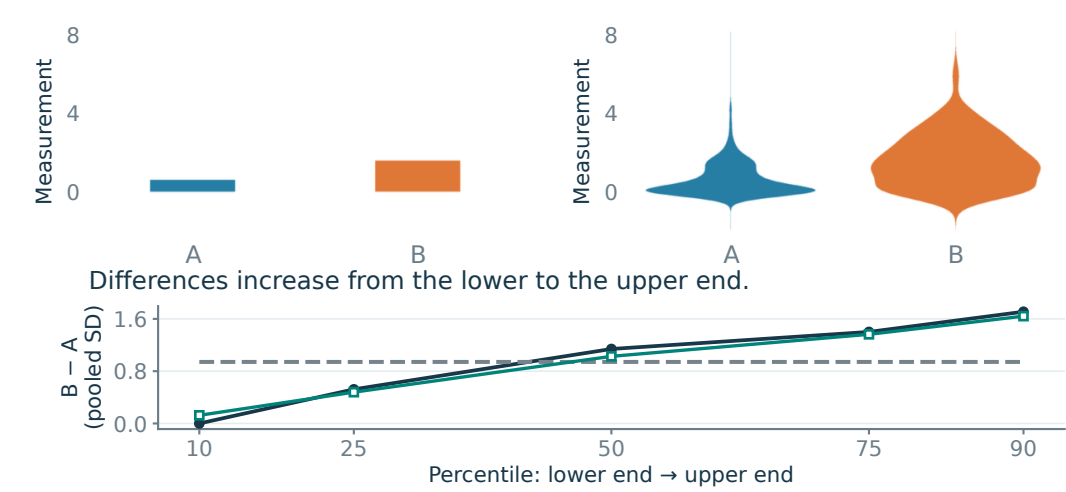
B: Disease (n=139)

Mean bars 64.22

→ Violin plot 92.84

+28.62

score points



RANK / CHART		SCORE / 100	
1	Violin plot 256 grid points		92.837
2	Histogram 64 bins		92.809
3	Heatmap 64 bins		92.014
4	Dot plot raw observations		91.860
5	Interval plot five-number		76.182
6	Box plot five-number		75.923
7	Bar chart means		64.220

When changing the chart adds little

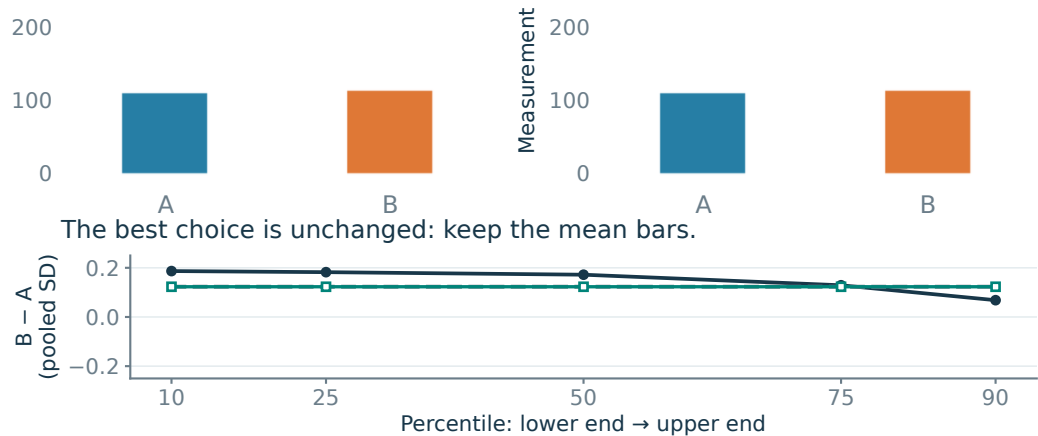
A high-scoring baseline leaves little room for improvement

True difference Recovered from mean bars Recovered from selected chart

a Wrist exercise / Mean heart rate

A: Early third (n=8) B: Late third (n=8)

Mean bars 95.35 → Bar chart 95.35



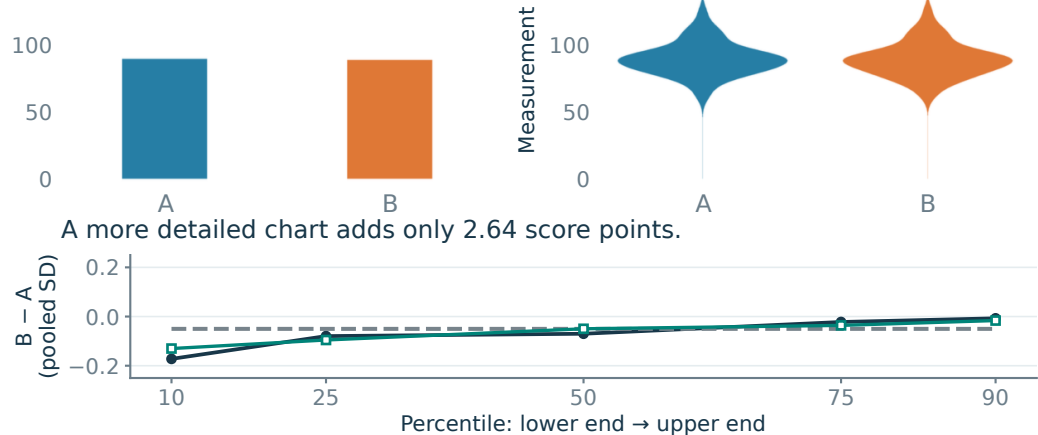
+0.00 score points

RANK / CHART		SCORE / 100	
1	Bar chart means	95.348	
2	Dot plot means	95.036	
3	Violin plot 16 grid points	93.980	
4	Histogram 8 bins	89.312	
5	Heatmap 8 bins	89.270	
6	Interval plot five-number	81.152	
7	Box plot five-number	80.946	

b Bedside monitoring / Mean heart rate

A: ECG HR (n=46) B: PPG pulse (n=46)

Mean bars 95.38 → Violin plot 98.02



+2.64 score points

RANK / CHART		SCORE / 100	
1	Violin plot 128 grid points	98.019	
2	Histogram 4 bins	97.330	
3	Heatmap 32 bins	97.036	
4	Dot plot raw observations	96.865	
5	Bar chart means	95.377	
6	Interval plot five-number	73.519	
7	Box plot five-number	73.407	